Keerthi Chirumamilla

Data Engineering | Data Analytics | Python & SQL
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PROFESSIONAL SUMMARY

Data-focused developer with hands-on project experience in SQL and PostgreSQL data modelling, REST API integration and data validation with schema checks and automated tests. Built and deployed two database-backed web applications (Supabase, Vercel) with authentication and Row Level Security. Skilled in Python data cleaning, exploratory analysis and Power BI / Tableau dashboards. Seeking Werkstudent, internship or junior roles in data engineering, data analytics and BI.

TECHNICAL SKILLS

Databases & Data Modelling: SQL, PostgreSQL, MySQL, MongoDB, relational schema design, SQLAlchemy ORM, versioned SQL migrations, Row Level Security policies, SQL security tests

Data Integration & Quality: REST API integration (Finnhub, Gemini), structured JSON output, request and response validation (Zod, Pydantic), rate limiting, input checks

Data Analysis & BI: Python, Pandas, NumPy, Matplotlib, exploratory data analysis, data cleaning and preprocessing, data visualisation, Power BI, Tableau, Excel

Machine Learning: Linear Regression, Random Forest, LSTM, deep learning, model evaluation, predictive forecasting

Programming: Python, SQL, TypeScript, JavaScript, Java, C

Backend & Web: FastAPI, Django, Node.js, Java EE (Servlets, JSP), Next.js, React, Vite, Tailwind CSS, JWT authentication

Cloud, DevOps & Tools: Supabase (PostgreSQL, Auth, Edge Functions), Vercel, AWS and Google Cloud (foundational certifications), Git, GitHub, Vitest

INTERNSHIP EXPERIENCE

AI-ML Virtual Internship – Eduskills (AWS Academy) 05/2023 – 07/2023

- Built machine learning and deep learning models in Python as part of a structured virtual programme.
- Applied data preprocessing and feature preparation to improve model inputs.
- Evaluated models with standard metrics and compared results to improve predictive accuracy.

Data Analytics Virtual Internship – Eduskills 12/2022 – 02/2023

- Cleaned, transformed and prepared datasets using Excel, SQL and Python.
- Performed exploratory data analysis to identify patterns and trends.
- Summarised findings as business-intelligence insights to support data-driven decisions.

KEY PROJECTS

Veyoraq – AI-Powered Personal Styling Platform (Personal project) 10/2026 – Present

Next.js, React, TypeScript, Supabase (PostgreSQL, Auth, RLS), Gemini API, Zod, Vitest, Vercel

Live: veyoraq.vercel.app | GitHub: github.com/chirumamillakeerthi22/Veyoraq

- Built and deployed a full-stack styling web application (style quiz, colour analysis, fit guidance and photo-based outfit advice) with server-side Next.js API routes.
- Integrated the Gemini API (text and image input) with structured JSON output; validated every request and AI response against Zod schemas so malformed data is rejected before it reaches the user.
- Designed PostgreSQL tables in Supabase for saved outfits, wishlists and colour reports, with versioned SQL migrations, Row Level Security and SQL test scripts that check per-user access, size limits and row caps.

- Hardened the API layer with same-origin checks, per-visitor rate limits, input size and file-type checks and server-side secrets; wrote 531 passing automated tests across 49 test files (Vitest).
- Implemented email/password and Google sign-in with Supabase Auth, so saved looks and colour reports are stored per user account.

InvestTrack – Stock Portfolio Platform (Personal project) 08/2026 – 09/2026

React, Vite, FastAPI, SQLAlchemy, PostgreSQL, Supabase (Auth, Edge Functions), JWT, Finnhub API, Recharts, Vercel

Live: invest-track-flame.vercel.app | GitHub: github.com/chirumamillakeerthi22/investTrack

- Integrated the Finnhub REST API (quotes, company search, price history) through three Supabase Edge Functions, keeping the API key server-side.
- Built holdings, watchlist and wishlist features with portfolio value and profit/loss calculations and price-history charts in a React dashboard (Recharts).
- Stored user portfolios in Supabase PostgreSQL and modelled users, stocks and holdings with SQLAlchemy using exact Numeric types; the FastAPI backend verifies Supabase-issued JWTs.
- Implemented registration, login, password reset and protected routes with Supabase Auth and React Router; deployed the front end on Vercel.

Air Quality and Pollution Forecasting (Academic capstone)

Python, Pandas, NumPy, Power BI, Tableau

- Forecast pollution levels by comparing Linear Regression, Random Forest and LSTM models, with data preparation in Pandas and NumPy.
- Visualised forecasts and AQI calculations in Power BI and Tableau dashboards.

ADDITIONAL PROJECTS

- **Hospital Management System** (Python, Django, MySQL): authentication, patient records, scheduling and billing modules.
- **Employee Management System** (Java EE, Servlets, JSP, MySQL): attendance, payroll and role-based access.

EDUCATION

M.Sc. Web and Data Science – University of Koblenz, Germany 10/2025 – Present

B.Tech, Computer Science Engineering – KL University, India (CGPA 9.35) 08/2021 – 05/2025

CERTIFICATIONS

Google Cloud Practitioner – Google Cloud 07/2024

AWS Cloud Practitioner – Amazon Web Services 12/2023

Red Hat Enterprise Application Development (Java EE) – Red Hat 08/2023

LANGUAGES

English – fluent | German – A2 (currently learning)